MILITARY SPECIFICATION

COUPLER, ATTITUDE INDICATOR TYPE CU 1466*/ASN

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide flight director information to the attitude indicator during bombing maneuvers, and additionally shall provide certain controls to the Attitude Heading Reference System and provide interface between the AHRS and other equipment.

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.11 of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4.)

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in WR-98.

3.2.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied as outlined in Specification MIL-E-4682.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the current version before use.

3.2.3 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of WR-98.

3.2.4 Modules, Maintenance - The electronic portions of the equipment shall be divided into maintenance modules (see Specification MIL-S-23603). Maintenance modules shall normally be considered repairable.

3.2.5 Microelectronic Interchangeability - The interchangeability provisions of WR-98 shall govern Microelectronic interchangeability.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 11.5 pounds.

3.3.2 Reliability

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 100 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 1700 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The equipment shall contain time totalizing meter in accordance with Specification MIL-M-7793. Meter shall be type MS17322.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Case - The case of the coupler shall conform to Figure 2 of this specification, have non-ferrous, low density metal mounting flanges, and shall be finished with a lusterless black material color No. 37038 of FED-STD-595.

3.3.5 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:35Z
Check the source to verify that this is the current version before use.

3.3.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0V AC and 27.5 ± 0.5V DC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein:

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve IV of Specification MIL-E-5400.

3.3.10.2 Acoustic Noise - The equipment shall operate satisfactorily when subjected to acoustic noise with a sound pressure level of 140 db.

3.3.10.3 Acceleration - The equipment shall operate satisfactorily when subjected to the tests of 4.2.2.2.

3.3.11 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 1.0 minute under standard conditions and 2.0 minutes at extreme service conditions.

3.3.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase) 115V, Category "B", 75 VA.
(2) DC Power 28V, Category "B", 0.8 Amp.

3.3.12.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.13 Dielectric Breakdown - The materials used and the design shall be such that the resistance between isolated circuits and between each circuit and the case or connector shell is at least 50 megohms when a potential of 500V DC is applied for a period of 10 seconds.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall provide interface between the AN/ASN-50 Reference System, the CP-741/A weapons release computer, and the ID-1329/A Attitude Indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

The unit provides control of the ASN-50 roll gimbal during LABS bombing maneuvers, and provides vertical and horizontal steering signals during all bombing modes. The unit provides bombing control interlocks and a tone output signal during pull-up in the bombing mode (see Figure 1).

3.5 Detail Requirements - The Attitude Indicator Coupler shall meet the following requirements:

3.5.1 Functions - The Attitude Indicator Coupler shall provide signals and switching associated with All-Attitude, LABS and Shrike modes of system operation as follows:

(1) Sum, amplify, and discriminate roll and yaw signals to provide a signal to drive the vertical pointer of an attitude-director indicator during LABS and Shrike modes of operation.

(2) Provide a signal to drive the horizontal pointer of an attitude-director indicator as a function of the difference between the acceleration sensed by an accelerometer and the desired acceleration represented by a signal programmed in the unit during LABS and Shrike modes of operation.

(3) Provide a continuous heading error signal during pull-up in the bombing modes based upon heading prior to pull-up.

(4) During Loft/OTS bombing maneuver, if aircraft is in a 90° climb and yaw attitude deviates 30°, the bombing mode shall be automatically cancelled and the system shall revert to its primary function of all-attitude indication.

(5) During Loft/OTS mode, provide signals for control of the displacement gyroscope outer roll gimbal by resolving inner roll and azimuth signals as a function of pitch, and using the resolved signal to control the outer roll gimbal. Reverts to all-attitude mode of operation upon loss of DC power.

(6) Provide 1200 cycle warning tones prior to and during pull-up in the LABS mode of operation as commanded by related equipment.

(7) Revert to all-attitude mode at Bomb release.

(8) Switch the signal to the Vertical Director pointer so as to hide the pointer from view when not in the bombing mode(s).

(9) Provide relay contacts to distribute an Attitude and Heading system failure warning.

(10) Cut off pitch erection during bomb modes for the period between pull-up and release.

3.5.2 Performance - The Attitude Indicator Coupler shall provide the following performance when properly connected.

3.5.2.1 Horizontal Director - The Attitude Indicator Coupler shall provide signals to the Horizontal Director Pointer which shall provide guidance for performing a constant "g" pull-up maneuver. It shall be adjustable to the center of the indicator and during the pull-up shall have a scale factor adjustable to 3/4 inch/g. The entry easement interval to sustain pull-up g's shall be 2 seconds.

3.5.2.2 Vertical Director - The Attitude Indicator Coupler shall provide signals to the Vertical Director pointer which shall provide guidance for maintaining proper roll and yaw attitude during a bombing maneuver. It shall have a scale factor adjustable to 1/16 inch/deg. for roll error indication

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

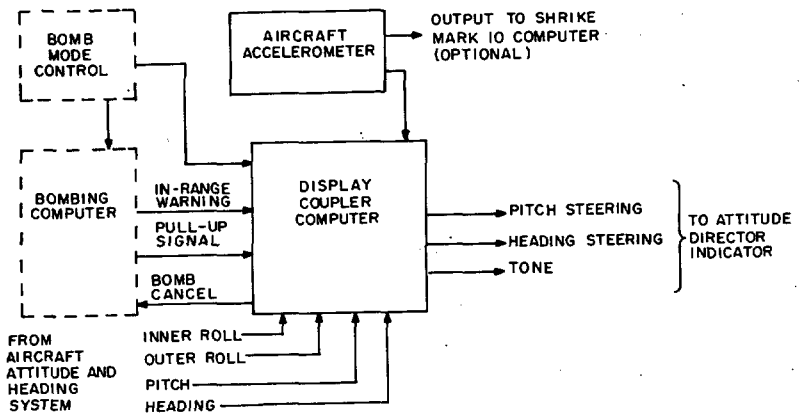


Figure 1. System Diagram

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

and 3/16 inch/deg. for yaw error indication. It shall deflect in the direction in which a corrective roll maneuver must be applied, e.g., deflection to left, left wing down roll maneuver is required.

3.5.2.3 Roll Tracking In Bombing Mode - In any bombing mode the display coupler shall provide signals to the sphere of the Attitude Indicator to provide complete all-attitude indication.

3.5.2.4 Roll Cancel - During Loft/OTS bombing mode, if aircraft is in a 90° climb and yaw attitude deviates 30°, the bombing mode shall be automatically cancelled and the system shall revert to its primary function of all-attitude indication.

3.5.3 Construction

3.5.3.1 Modules - The Attitude Indicator Coupler functions listed shall be modularized for maintenance purposes:

(1) "G" Programmer
(2) Amplifier-demodulator, Yaw-Roll
(3) Power Supply and Tone Generator
(4) Heading Synchronizer
(5) Pitch Resolution module
(6) Relay Modules (3)

3.5.3.2 Adjustments - The equipment shall contain adjustment potentiometers as necessary. Potentiometers shall be contained within its associated module.

3.5.3.2.1 External adjustments - The following adjustments shall be accessible from the exterior of the unit. They shall be protected to prevent accidental movement.

(1) Horizontal Director Centering (1g Adjustment)
(2) Roll Sensitivity
(3) Yaw Sensitivity
(4) G Error Sensitivity
(5) Bombing Tone Level
(6) 1-4g Run Down Time

3.5.3.3 "G" Programmer - The "G" Programmer shall develop a DC output signal whose amplitude is directly proportional to the difference between the "G" force applied to an external accelerometer and an internally programmed signal representing a desired "G" load. The output shall be capable of driving two 1000 ohm resistive loads such that 1/2 "G" error shall produce 0.94 volts across the load. The program shall have a start and a stop voltage equivalent to that developed by the accelerometer under a 1 and 4 "g" acceleration. The 3 "g" voltage change shall be adjustable to 2 seconds duration by the "Prog" adjust potentiometer.

3.5.3.3.1 Alternate "G" Program - When commanded by both of two external signals (+28V DC) corresponding to (1) either of two bombing modes, and (2) weapon type, the "g" programmer shall produce a stop voltage equivalent to 2.5 "g".

3.5.3.4 Roll-Yaw Module - The Roll/Yaw module shall provide signals in the form of a variable DC Voltage with reversible polarity which represents a total of roll and yaw input signals. The output signal shall be capable of driving two 1000 ohm resistive loads. The yaw input shall be from the resolver of the pitch resolver section. The roll input shall be derived from the outer roll synchro in the displacement gyroscope. Externally accessible adjustments shall be provided to adjust roll and yaw sensitivities.

3.5.3.5 Power Supply and Tone Generator - The power supply shall provide AC and DC outputs to the other modules within the coupler as required to perform all functions specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.5.3.5.1 Tone Generator - A 1200 cycle tone output signal shall be provided when the pull-up voltage or tone warning voltage is applied. The magnitude of the tone signal shall be adjustable from null to 1.6 volts across a 280 ohm load. It shall be contained within the power supply module.

3.5.3.5.2 Pull-Up Relay - A relay shall be contained within the power supply module which is operated upon pull-up. One set of contacts operates in the azimuth lock circuit and the other provides pitch erection cutout.

3.5.3.6 Heading Synchronizer - The Heading Synchronizer module shall consist of a servo mechanism which positions a differential transmitter in response to the information received from the heading reference system. The synchronizer shall develop a signal proportional to aircraft yaw upon application of a pull-up signal which stops the servo motor. The servo loop shall have a null sensitivity of ± 0.25° maximum. The servo loop shall follow a turn rate of 50° per second minimum.

3.5.3.7 Pitch Resolver Module - The Pitch Resolver Module shall consist of a servo mechanizm which positions a resolver in response to aircraft pitch attitude information received from a pitch synchro transmitter (displacement gyroscope). The output of the resolver is supplied to the Roll/Yaw section and the Roll pre-amplifier section. The resolver shall not lag the transmitter by more than 3.4° when the transmitter is rotated at a rate of 50° per second and shall not lag by more than 0.7° at a rate of 12° per second. There shall be perceptible movement of the resolver when the transmitter is displaced 0.25° from the reference point. The resolver shall be at a null when the transmitter is at 0° ± .5°.

3.5.3.8 Relay Module No. 1 - This relay module shall contain relays for switching synchro signals and a power failure relay. The module shall contain relays which, when energized by external sources (ground) function to reverse the direction of rotation of the servo mechanism and reference signals for a given direction of rotation of the input signals. The power failure relay shall accept a command signal from the AN/ASN-50 and provide indicator flag operation and autopilot cutout signals.

3.5.3.9 Relay Module No. 2 - This relay module shall contain all relays necessary to the control of the AN/ASN-50 third gimbal during bombing modes. It shall also provide command signals for the proper phasing of the vertical pointer director signals.

3.5.3.10 Relay Module No. 3 - This relay module shall contain relays associated with control of the bombing mode activation and bomb release interlocks. It shall also contain the yaw cancel circuits.

3.5.3.10.1 Yaw Cancel - Inner roll error signals shall cause the relay switching to revert to all attitude operation. The angle at which cancellation occurs shall be adjustable from 25° to 45°.

3.5.3.11 Dimensions and Mounting - The mounting and overall dimensions of the unit shall be in accordance with Figure 2. The equipment shall be suitable for mounting in any attitude.

3.5.3.12 Electrical Connections - Electrical connectors shall be mated with Type MS 3116-18-32SW and MS 3116-22-55S, and shall be connected in accordance with Figure 3.

3.6 Associated Equipment - The coupler shall operate with the associated equipment listed in 6.9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

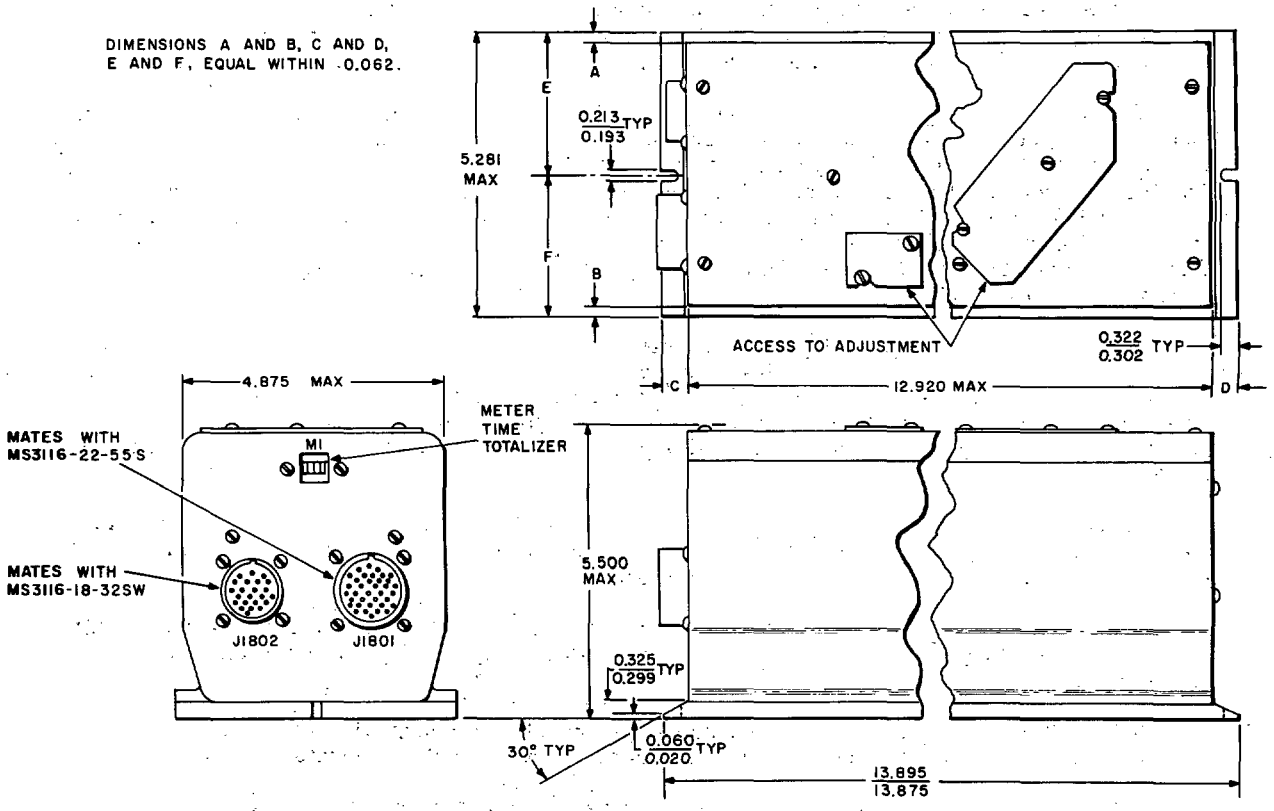


Figure 2. Outline Drawing

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

Connector J1801 Mates with MS 3116-22-55S			Connector J1802 Mates with MS 3116-18-32SW		
FUNCTION		TERMINAL	FUNCTION		TERMINAL
Pow. Fail. Relay (+28 T.D.)		A	Cage, 28V DC, Rect.		A
Pow. Fail. Relay (A/P Use)		B	Horiz. Pointer		B
90° Pitch Switch		C	Acceler. Excit.	(C)	C
90° Roll Switch		D	Reserved		D
Pull-Up		E	Tone Gen. Out		E
Loft - O.T.S.		F	Horiz. Pointer		F
Bomb Release Com'd.		G	Acceler. Input		G
Inner Roll CX - (X)		H	Pitch CX Rotor		H
Bomb Released Sig.		J	Vertical Pointer		(J
Roll Gen.		K			(K
Cage at Pull-Up (Reserved)		L	Cage Relay		L
Stick-Toss Mode		M	Roll & Pch. CX Excit.	(H)	(M
+28V DC		N		(C)	(N
115V AC, -Phase B		P	Pitch CX Rotor		P
115V AC, -Phase C		R	Pitch CX Stator Out		R
Power, Neutral		S	Pitch CX Stator In		S
Azim. Reverse Relay		T	Pitch CX Stator Out		T
Pow. Fail. Relay (A/P Use)		U	Pitch CX Stator In		U
Pow. Flag. Common		V	Spares		V, W
	(X	W	Pitch Erect. C.O.		X
Azim. Ref. CX	(Y	X	Roll CX Rotor		Y
	(Z	Y	Spares		Z, a
Spare		Z	Vertical Pointer		b
	(X	a	Horiz. Pointer		c
Pitch Ref. CX	(Y	b	Roll CX Rotor		d
	(Z	c	Acceler. Excit.	(H)	e
Outer Roll CX	(Y	d	Pitch CX Stator In		f
	(X	e	Pitch CX Stator Out		g
Pickle, +28V DC		f	Spares		h, j
	(	g			
Azim. Servo	(	h			
Phase Rev.	(	i			
Tone Warning		j			
Shrike Weapon		k			
AZ. Servo Phase Rev.		m			
Pow. Fail. Relay (A/P)		n			
Roll Servo Input		p			
Inner Roll CX	(Y)	v			
Spares		q thru u			
Spares		w thru HH			

Figure 3. Connector Wiring

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:35Z
Check the source to verify that this is the current version before use.

specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422, except as modified herein and in 3.3.10, and interference tests in accordance with Specification MIL-I-6181.

4.2.2.1 Acoustical Noise - The equipment shall be subjected to the noise test of method 515 of MIL-STD-810, Procedure 1 to grade level A.

4.2.2.2 Acceleration - The equipment shall be subjected to the acceleration tests of method 513 of MIL-STD-810 for Aircraft and Helicopters. During the acceleration of Procedure II, the equipment shall operate satisfactorily. After exposure, the equipment shall meet the requirements of the individual tests.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operation Test
(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests. Tests shall include Temperature Tests steps 2 and 6 of MIL-T-5422 (Para. 4.1.2) and Vibration Test step 3, Part I of MIL-T-5422 (Para. 4.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. A Qualification Phase and Sampling Phase Tests shall be conducted. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the Section entitled "Sampling Phase of Production Reliability Tests". The Accept-Reject criteria for Test Plan IV shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot Size for Sampling Phase - The equipments constructed during one month shall be one lot, except minimum lot size shall be 12.

4.4.3.3 Test Level - Except for the initial cycle of 4.4.3.3.1 the test level shall be in accordance with Level E of MIL-STD-781.

4.4.3.3.1 Initial Cycle - Upon receipt of the equipment, following operational tests (or debugging), the coupler shall be placed in a chamber at +71°C (+160°F) and operated for 4 hours using nominal voltage. The coupler shall then be tested at -54°C (-65°F) after a 4-hour soak at -54°C (-65°F). The initial temperature cycle shall be conducted at the start of each sample lot and on each unit upon repair or replacement.

4.4.3.4 Duty Cycle - The coupler shall be operated for 2-1/2 hours at nominal voltage and shall be de-energized for 1/2 hour. This duty cycle shall be conducted at least eight times each day. Time counted toward the reliability test shall be the sum of all 2-1/2 hour running periods.

Following the initial hot and cold tests of the one temperature cycle (see 4.4.3.3.1), the tests shall continue at the test level of para. 4.4.3.3.

4.4.3.5 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified in 6.10.

4.4.3.6 Failure Criteria - In addition to the requirements of Specification MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the Acceptance requirement (paragraph 4.4.3.5) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted in equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	27.0 ± 2.0 volts

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full par-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

ticulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The coupler covered by this specification is intended for use in aircraft to provide signals and indication for a bomb delivery system. Provisions are made for various types of operation and error compensation depending upon the application and associated equipment.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5V DC and 115 ± 1.0V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Reference Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.
(2) Selection of applicable levels of packaging and packing (see 5.1).
(3) Data requirements (see 3.6).

6.8 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.9 Associated Equipment -

Item	Type Designation
Transmitter, Remote Compass	ML-1
Gyroscope Assembly, Displ.	CN-1054/A
Amplifier-Power Supply	AM-4274/ASN-50
Adapter-Compensator, Compass	MX-6985/ASN-50
Controller, Compass System	C-2984/AJB-3A
Gyroscope, Rate Switching	CN-495A/AJB-3
Indicator, Attitude Director	ID-1329/A
Computer, Weapons Release	CP-741/A
Accelerometer, Aircraft	MX-2911/AJB-3A

6.10 Performance Characteristics to be Measured - Tests necessary to determine failures in performance characteristics (listed below) shall be conducted at least once every 24 hours. If operation is extended over a weekend, the tests may be conducted on the day prior to and the day following the weekend in lieu of the 24-hour tests for the weekend.

Function	Accept Performance
Switching Functions (Pickle, Pull-Up & Bomb Release)	Functional Check
Vert. Pointer Operations (Roll, Pitch, Azimuth)	3/8 ± 1/8 inches per 2° azimuth 1/2 ± 1/8 inches per 8° outer roll
Horiz. Pointer	3/8 ± 1/8 inches per 1/2 g error sensitivity 2 ± 1/4 seconds for 1 to 4 g rundown
Bomb Cancel	30° + 15°, -5° (to be set)
Tone Generator	1.6 ± 0.6V across 280 ohms

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:35Z
Check the source to verify that this is the currentversion before use.

6.11 Applicable Documents -

SPECIFICATIONS

Military

MIL-E-4682	Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MS17322	Meter, Time, Totalizing, Miniature 400 Cycle, Digital

6.11.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-81446(AS) COUPLER, ATTITUDE INDICATOR TYPE CU 1466*/ASN

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

Naval Air Systems Command
Washington, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

NAVAL AIR SYSTEMS COMMAND(AIR-52021)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the currentversion before use.